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ORIGINAL ARTICLE

Fabrication of *Achyranthes aspera* extract loaded ZIF-8 nanocomposite for lung cancer treatment, an in vitro assessment

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Abstract

Medicinal plant-based biocompatible nanomaterials have become increasingly important in the healthcare field. Zeolitic imidazole (ZIF-8) frameworks are fascinating porous crystalline materials that have great attention in the field of drug delivery due to their biodegradability, pH sensitivity to changes in acidic conditions, and chemical durability. The development of ZIF-8 particles loaded with *Achyranthes aspera* extract (AA) and the evaluation of their anticancer properties are the primary findings of this study. Various spectroscopy and macroscopic strategies were utilized to completely investigate the material characteristics of the AA@ZIF-8 nanocomposite. MTT results of the AA@ZIF-8 nanocomposite demonstrated significant anticancer potential against A549 cells with an IC_{50} value of 63.22 ± 0.03 μg/mL. Fluorescent microscopy investigations have demonstrated that enhanced levels of reactive oxygen species (ROS) can lead to DNA damage, which could be the possible cause for the cytotoxic effects of an AA@ZIF-L nanocomposite on the A549 cells. The materials' biocompatibility has been proven via the *Artemia salina* acute toxicity test. The *Artemia salina* acute toxicity assay results confirmed that the nanocomposites' LC_{50} value was 160.13 ± 2.82 μg/mL, with a nontoxic nature. According to the research findings, AA@ZIF-8 nanocomposite is a promising biocompatible material for drug delivery system for lung cancer therapy and other biomedical applications.

Keywords Zeolitic imidazole frameworks (ZIFs) · Medicinal Plants · *Achyranthes aspera* (AA) · Anticancer activity · Biocompatibility

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Introduction

Zeolitic imidazole frameworks (ZIFs) are popular among biomedical researchers because of their enormous surface area, variable porosity structure, and diverse topologies (Biswal et al. 2013). ZIFs are an ideal platform for drug administration due to their chemical and thermal stability, high drug-loading capacity, pH-sensitive degradation, regulated drug release, and biocompatibility (Adhikari et al. 2015; James and Lin 2016). Plant-based therapies have limitations, including poor solubility, non-targeted drug delivery, short half-life, heat instability, and immunological clearance (James and Lin 2016). Encapsulating plant-derived chemicals or phytocompounds within ZIFs can address these limitations (Tiwari et al. 2017). ZIFs' greater stability in aqueous environments prevents early degradation of phytocompounds, improves intracellular absorption, and enables targeted medication administration without harming healthy tissue (Jones et al. 2016; Zhang et al. 2017). This innovative technique addresses the limitations of Ayurvedic therapy, which uses plant-based remedies to treat a wide range of diseases (Jones et al. 2016). Nanomaterials made from plant extracts are more stable, environmentally benign, and cost-effective (Zhang et al. 2017). *Achyranthes aspera* L. (family: Amaranthaceae) is a common medicinal plant found throughout tropical Asia. Herbal medicines were the sole means of curing ailments earlier than the invention of modern medicines. Nearly eighty percent of the global population, who inhabit the vast rural regions of developing and underdeveloped countries, is expected to remain mainly dependent on herbal medicines (Malvezzi et al. 2023). This herb plays a major role in Chinese medicine (CM) practices and Indian Ayurvedic (IA) treatments. Its hepatoprotective, anti-inflammatory, antitussive, and antifungal effects have been documented. Its antibacterial and wound-healing qualities are also utilized (Xia et al. 2022; Leiter et al. 2023; Zang and Wynder 1996; Khaltaev and Axelrod 2020). A large spectrum of phytocompounds from plants, such as cardiac glycosides, tannins, polyphenols, alkaloids, flavonoids, terpenoids, and essential oils, has been used to prevent a variety of diseases such as cancer, ulcers, wounds, and diabetes (Lakshmanan et al. 2018; Praveena and Kumar 2014).

The bulk of cancer-related deaths are triggered by lung cancer, an incurable neoplastic disease with the highest incidence of morbidity and mortality. Lung cancer continues to be among the major causes of cancer-related mortality, with nearly 1.8 million deaths (18%) in 2020, according to the International Agency for Research on Cancer (IARC). Lung cancer makes up 5.9% of the total incidence of cancer and 8.1% of fatalities due to cancer in India (Siegel et al. 2024, Mathur et al. 2020). Around eighty percent of people with lung cancer are smokers. The primary disease factor of lung cancer is cigarette smoking, even though it may also affect non-smokers. Current NSCLC strategies include postoperative treatment and multimodal therapy such as chemotherapy, surgery, and radiotherapy (Wolf et al. 2024; Tamta et al. 2022; Tiwari et al. 2018). Chemotherapeutic agents exhibit low efficiency due to the immunosuppressive mechanism of cancer cells, which is the major obstacle in developing a nanodrug strategy. The resistance of tumor cells can overcome the anticancer efficacy of commercial chemotherapeutics (Srivastav et al. 2011; Sharma and Chaudhary 2015). Due to a lack of public understanding, insufficient healthcare facilities, and limited funds, the situation could potentially become worse. Plant compounds have recently received attention as an advantageous resource for the synthesis of nanostructures with enhanced anticancer capabilities (Ha et al. 2024; Joshi et al. 2023). Presently, limited reports are available on the fabrication of metal–organic nanocomposites using plant sources (Chai et al. 2024). Metal–organic frameworks (MOFs), multifunctional materials, are used in agriculture, bioimaging, and drug delivery applications (Tong et al. 2013; Abdelhamid et al. 2017; Raju et al. 2023). The excellent thermal stability, drug loading capacity, pH sensitivity, and biosafety nature of MOFs make it a suitable material for cancer treatments (Vodyashkin et al. 2023; Sun et al. 2012; Moharramnejad et al. 2023; Abdelhamid 2021; Matusiak et al. 2023; Zhang et al. 2023; Parsaei and Akhbari 2022; Prabhu et al. 2019; Deepika et al. 2019; Rajabi et al. 2015; Dobrovolskaia et al. 2008; Rahimi et al. 2024). MOF-based drug delivery devices are safe for healthy cells and selectively target cancer cells (Adhikari et al. 2015; James and Lin 2016). Zeolitic imidazole framework (ZIF-L) is a nontoxic and biodegradable metal–organic framework composed of acidic zinc ions and basic imidazole ligands. ZIF-L-based drug delivery systems have been proposed as an alternative to harmful nanoparticles in biomedicine (Tiwari et al. 2017; Zhang et al. 2017, 2023). In this work, we fabricated *Achyranthes aspera* extract (AA) loaded AA@ ZIF-8 nanocomposite in a simple method. The anticancer and biocompatibility of the nanocomposite have been examined against A549 lung cancer cells and L929 skin fibroblast cells. In the future, it will be a low-cost and biosafe therapeutic for lung cancer treatments after clinical trials.

Materials and methods

Chemicals used

Bovine serum albumin, Zinc Nitrate Hexahydrate, Penicillin, Streptomycin, and 2-methylimidazole (Himedia, India). The stains DAPI, MTT, and DMEM High-Glucose Medium were obtained from Sigma Aldrich. Human lung cancer cells

(A549) and L929 fibroblast cells were purchased from, National Centre for Cell Science (NCCS), India.

Preparation of plant extract

Fresh *Achyranthes aspera* leaves (AA) were collected from the Alagappa University, Science Campus, Karaikudi, Tamil Nadu, India. About 20 g of leaves was uniformly ground with a granite mortar. The leaf extract was collected and freeze-dried using a lyophilizer. The crystalline extract powder was stored in an airtight 5 mL storage bottle under cold conditions (– 20 °C) until further use (Raju et al. 2023).

Synthesis of phytoextract-loaded ZIF-8 nanocomposite

The *A. aspera* extract-loaded nanocomposite was fabricated using the previous methodology (Abdelhamid et al. 2017). Briefly, zinc nitrate 100 mg was dissolved in 25 mL of double distilled water. The 300 mg of 2-methylimidazole dissolved in 25 mL of DD water. The solution of 2-methyl imidazole was added to the zinc solution under magnetic stirring for 30 min. The color changes of the precursor to a white colloid confirm the zeolitic imidazole formation (ZIF). Further, 10 mL of *Achyranthes aspera* extract solution (2 mg/mL) was slowly added into the ZIF colloid, and the stirring was continued for the next 30 min. Finally, the formation of a light green colored colloid indicates the loading of the extract within ZIF nanoparticles. The colloid was washed twice by centrifugation at 10,000 rpm for 15 min. The hydrated colloid was dried for 6 h at 60 °C using a hot air oven.

Characterization of AA@ZIF-8 nanocomposite

Powder X-ray diffraction spectroscopy (XRD) was used to examine the crystalline nature and size of produced ZIF-8 and AA@ZIF-8. The surface plasma resonance of the AA@ZIF-8 nanocomposite was examined using a UV–Vis spectrophotometer (UV-3600 Shimadzu, Japan), and the characteristic absorption spectrum was determined using a wavelength scan in the 200–800 nm range. The functional group of AA@ZIF-8 was studied using FTIR spectral analysis in wave numbers ranging from 4000 to 400 cm^{-1}. Field Emission-Scanning Electron Microscopy (FE-SEM) (Carl Zeiss, Germany) was used to examine the surface morphological properties of the nanocomposite. The crystal morphology and size of AA@ZIF-8 were investigated using High Resolution-Transmission Electron Microscopy (HR-TEM) (FEI, Netherlands, model number Tecnai, G2 F20). The particle size of the produced AA@ZIF-8 nanocomposite was determined by dynamic light scattering (DLS) analysis (Micrometrics, Nano Plus) (Prabhu et al. 2019).

Drug loading efficiency

The extract loading efficiency of the synthesized AA@ZIF-8 nanocomposite was quantified by the previous method (Rahman et al. 2025). The concentration of free-floating leaf extract was calculated from the supernatant after the nanocomposite separation. Separation was performed under a cooling centrifuge for 20 min at 4 °C at under10,000 rpm. The supernatant was filtered with a membrane filter (0.22 mm). The percentage of extract loading was quantified at the wavelength of 270 nm using UV–vis spectroscopy (Perkin Elmer UV-spectrometer). The total percentage of drug loading was quantified by this formula:

$$\%\text{ of DLE} = \frac{\text{weight of the drug in NPs}}{\text{weight of the nanoparticle}} \times 100$$

Drug release study

Drug release kinetics were carried out in phosphate-buffered saline at pH-5, 6, and 7.4. The extract-containing AA@ZIF-8 nanocomposite (50 mg) was mixed into PBS and shaken at 37 °C at 100 rpm. At specified intervals, 2 mL of supernatant solution was taken and replaced with the same volume of the new buffer saline. The concentration of ciprofloxacin in the samples was determined using a UV–Vis Spectrophotometer (Shimadzu, 1700) at an absorbance range of 270 nm.

MTT assay

The cytotoxic effect of AA@ZIF-8 nanocomposite against lung cancer cells (A549) was investigated by MTT assay (Deepika et al. 2019). The cells were treated for 48 h with different concentrations of AA@ZIF-8 nanocomposite (10–100 µg/mL). After incubation, cells were treated with MTT (50 µL) at 37 °C for 4 h. Finally, the excess stain was removed by washing with 100 µL of DMSO solution. The characteristic absorbance was recorded using an ELISA multiplate reader (Softmax Pro V5 5.4.1 software, Molecular Device Spectramax M3) at 570 nm wavelength.

JC-1 staining for mitochondrial membrane potential loss

The damage of A549 cell mitochondrial transmembrane potential ($\Delta\Psi$m) was examined by the JC-1 staining technique (Raju et al. 2023). Following the experiment, A549

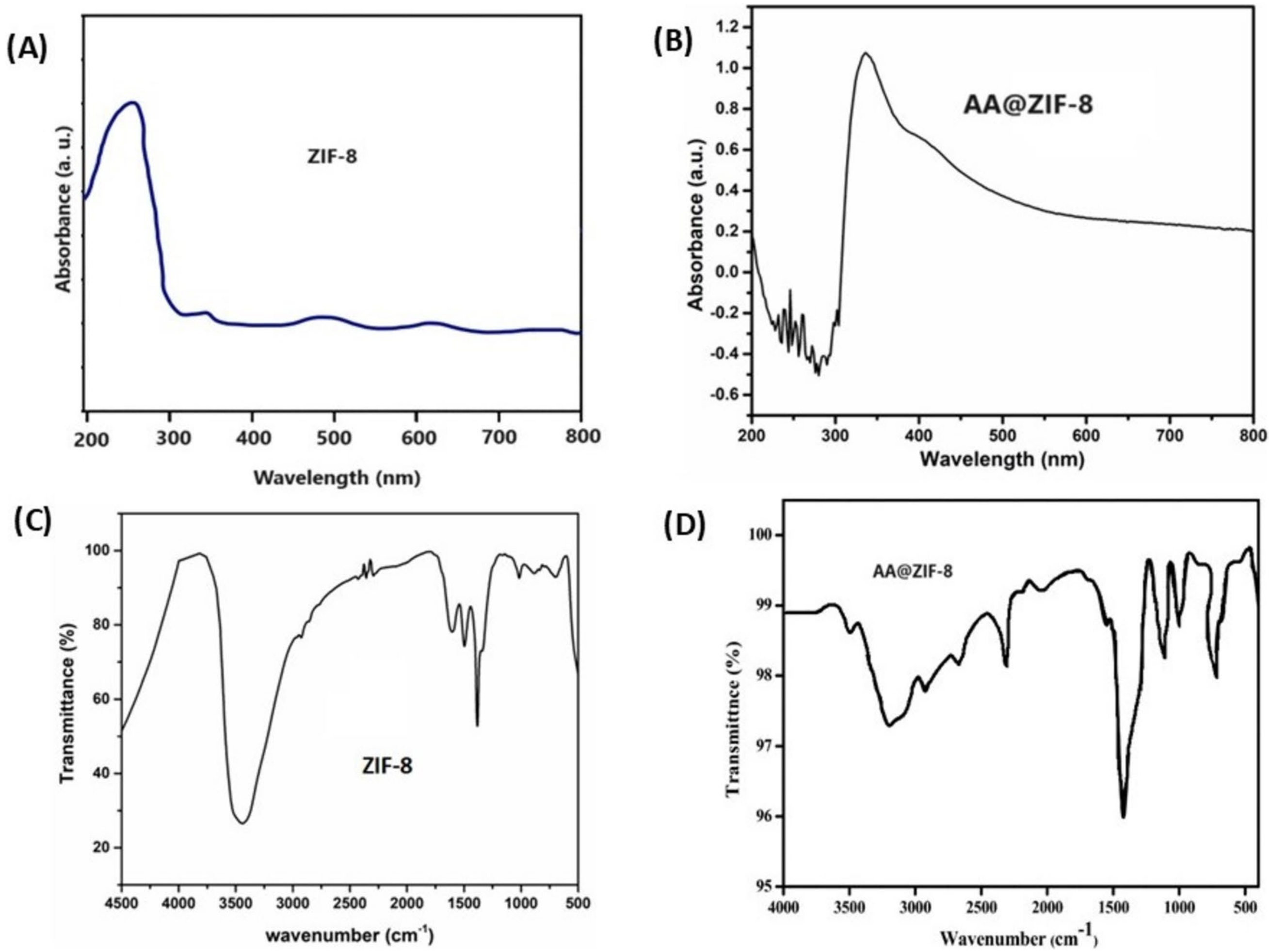

Fig. 1 **A**-**B** UV–vis spectra of ZIF-8 and AA@ZIF-8 nanocomposite, **C**-**D** FTIR spectra of ZIF-8 and AA@ZIF-8 nanocomposite

cells were treated with AA@ZIF-8 nanocomposite and AA leaf extract. After the post-treatment, cells were incubated at 37 °C for 30 min with 1 µg/mL of JC-1 stain. The excess JC-1 stain was removed by washing with PBS solution. Fluorescent absorbance intensity was recorded by a microplate reader (Spectra Max) at 488/525 nm (excitation/emission). The reduction of membrane potential was quantified from the absorbance of treated cells. The images were taken by fluorescence microscopy (Nikon ECLIPSE, Japan) at 20 × magnification.

Assessment of caspase 3 activity of AA@ZIF-8 nanocomposite

A quantitative caspase enzymatic activity of AA@ZIF-8 nanocomposite was assessed according to the manufacturer's instructions for the colorimetric assay kit (Abcam, Germany). After the 24-h treatment of AA@ZIF-8 nanocomposite (100 µg/mL), cells were washed with PBS solution and analyzed for total protein by the Bradford assay (Tan et al. 2024). Samples containing 200 µg of total protein were assayed for caspase-3/9 activity with DEVD-pNA as a caspase-3-specific substrate. Absorbance was measured at 405 nm in a multiplate reader.

Hoechst staining for assessment of nuclear damage

The cellular nuclear changes and apoptotic mechanism of AA@ZIF-8 nanocomposite were evaluated by Hoechst 33,258 staining (Prabhu et al. 2019). Human lung cancer MCF-7 cells (1×10^3) were treated with leaf extract and AA@ZIF-8 at the concentration of IC_{50} for 48 h. Cells without nanocomposite were taken as a negative control (0.1% DMSO) solution, and cisplatin was considered as a positive control. After the treatment cells were cleaned with PBS. Further, 10 µg/mL of Hoechst 33,258 stain was added and again incubated at 37 °C for 30 min under dark conditions. Finally, the excess stains were washed thrice with PBS. Cell morphological modifications were observed with an excitation of 350/460 nm under

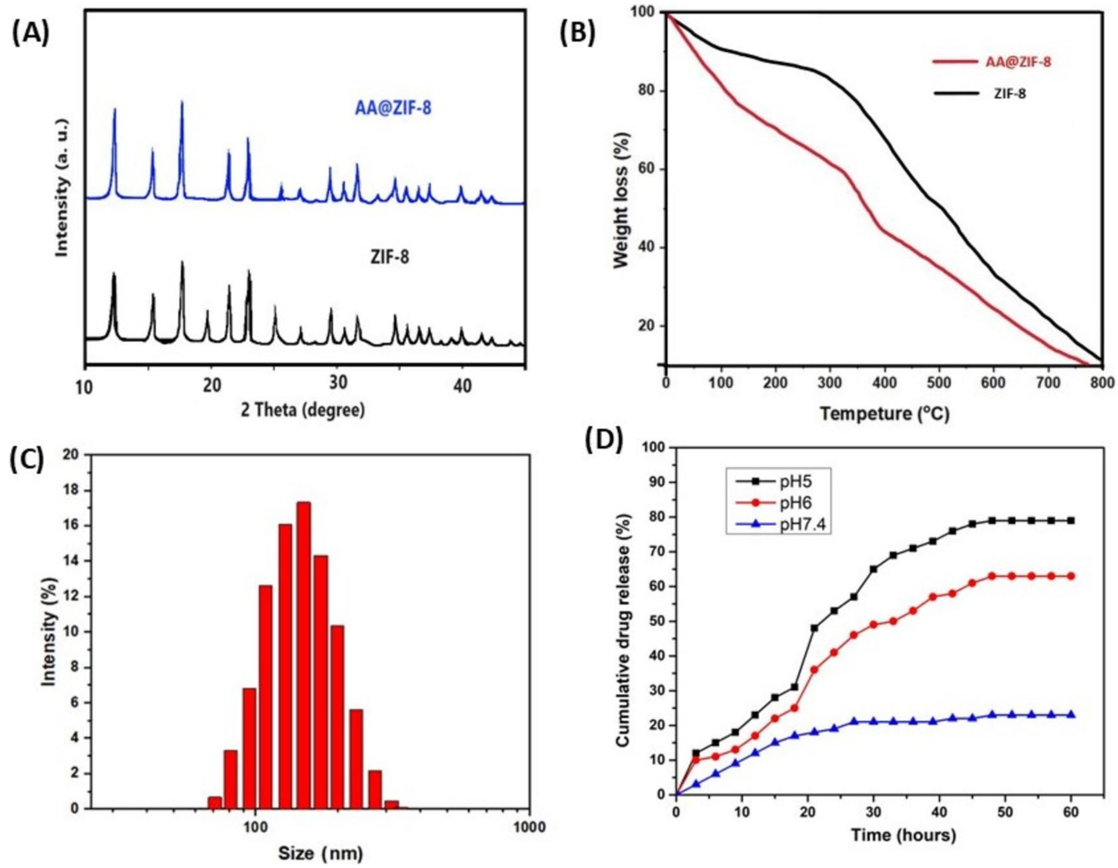

Fig. 2 **A** Powder XRD spectra of ZIF-8 and AA@ZIF-8 nanocomposite, **B** TGA spectra of ZIF-8 and AA@ZIF-8, **C** DLS particle analysis, **D** in vitro pH-responsive drug release mechanism of AA@ZIF-8 in different pH conditions

a fluorescent microscope. The percentage of apoptotic cells and nuclear membrane damage was investigated in five individual fields.

Assessment of apoptotic effects by AO/Et Br staining

The cellular change of apoptotic cells was evaluated by the AO/EtBr staining technique (Prabhu et al. 2019). Cells were treated

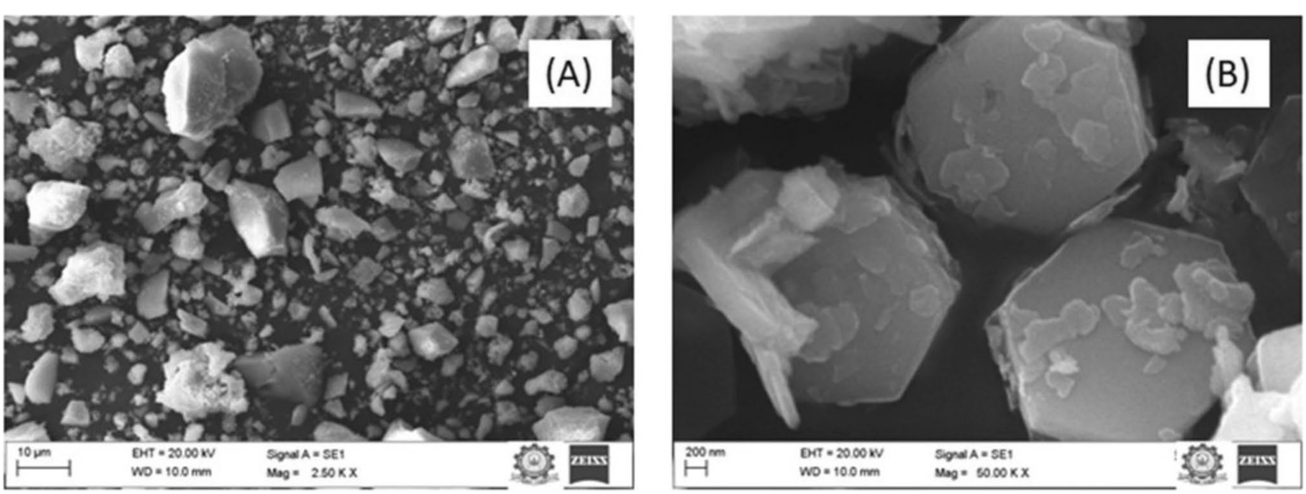

Fig. 3 **A**-**B** Scanning electron microscopic images of AA@ZIF-8 nanocomposite

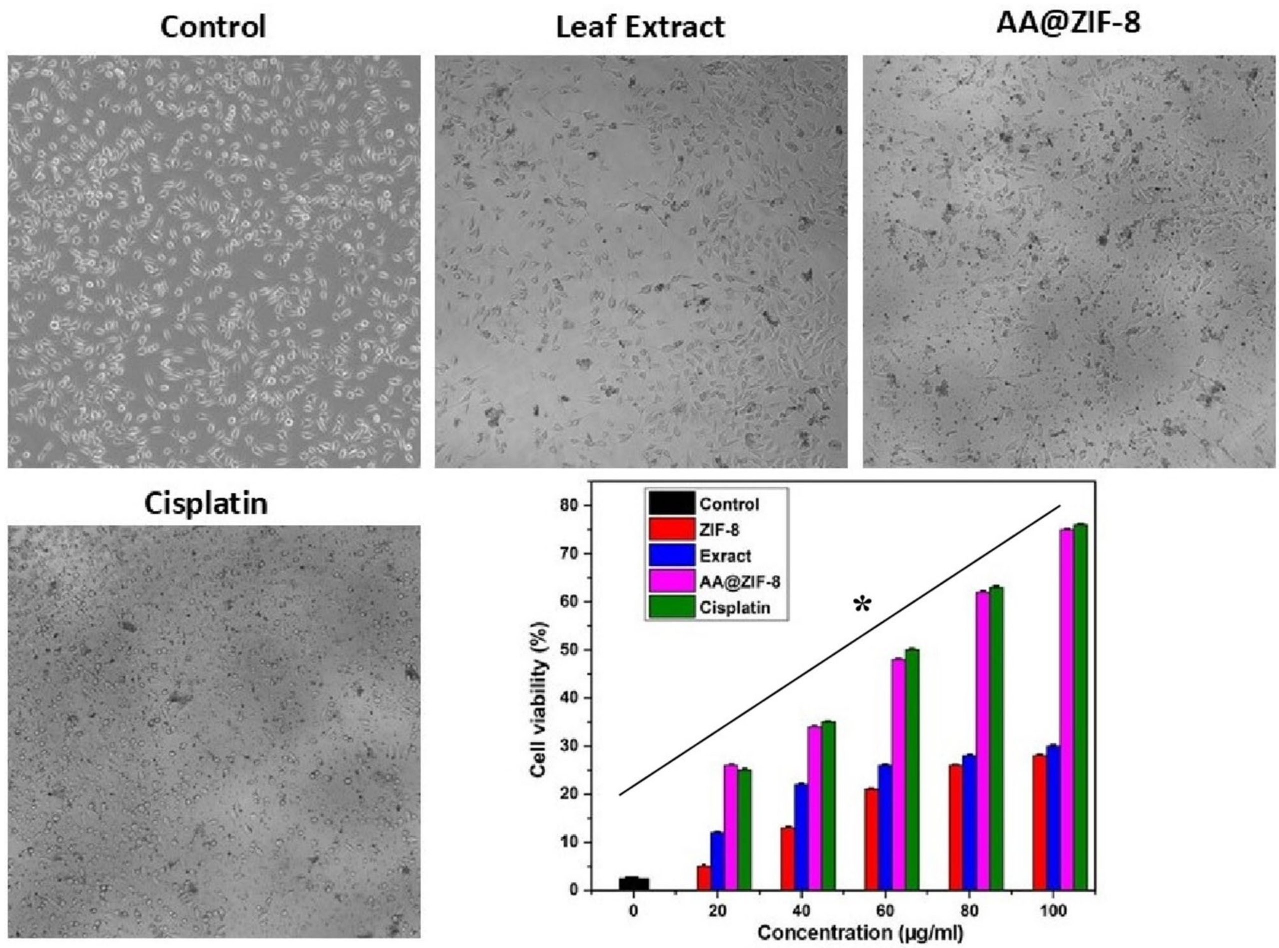

Fig. 4 Cytotoxic effect of AA@ZIF-8 against A549 lung cancer cells on MTT assay, microscopic images indicate the cell density reduction and cell damages under phase contrast microscopy. Results are expressed as mean ± SD of triplicate experiments. $*p < 0.05$ denotes the statistical significance between the control and treated

with AA@ZIF-8 nanocomposite and AA leaf extract at IC_{50} concentration at 37 °C for 24 h and fixed with 4% paraformaldehyde for 30 min. Finally, treated cells were subjected to AO/EtBr staining for 10 min, and changes were observed under a fluorescent microscope at a wavelength of 350/460 nm.

Acute toxicity assay

Cytotoxicity was evaluated (Rajabi et al. 2015; Dobrovolskaia et al. 2008) by *Artemia salina* (AS) lethality assay at different concentrations of AA@ZIF-8 under intermittent flow-through conditions. Around 20 nauplii were placed in 24-well plates containing 1 mL saline water with various doses of AA@ZIF-8 nanocomposite (25–150 µg/mL). Well-containing nauplii without a nanocomposite was considered as the control. The experiments were carried out in triplicate. The number of surviving nauplii was counted using a stereoscopic microscope. The mortality rate was quantified by comparison of the number of survivors in the control/treated groups using Abbott's formula.

$$\%\ \text{Mortality} = (\text{Number of deaths AS nauplii}) / (\text{Total number of live AS nauplii}) \times 100$$

Statistical analysis

The IC_{50}, $LD_{50,}$ and LD_{90} values have been calculated using Probit analysis software. Variance in control/treated groups was quantified by using ANOVA software. The * p-value $^{>}$ 0.05 was considered significant.

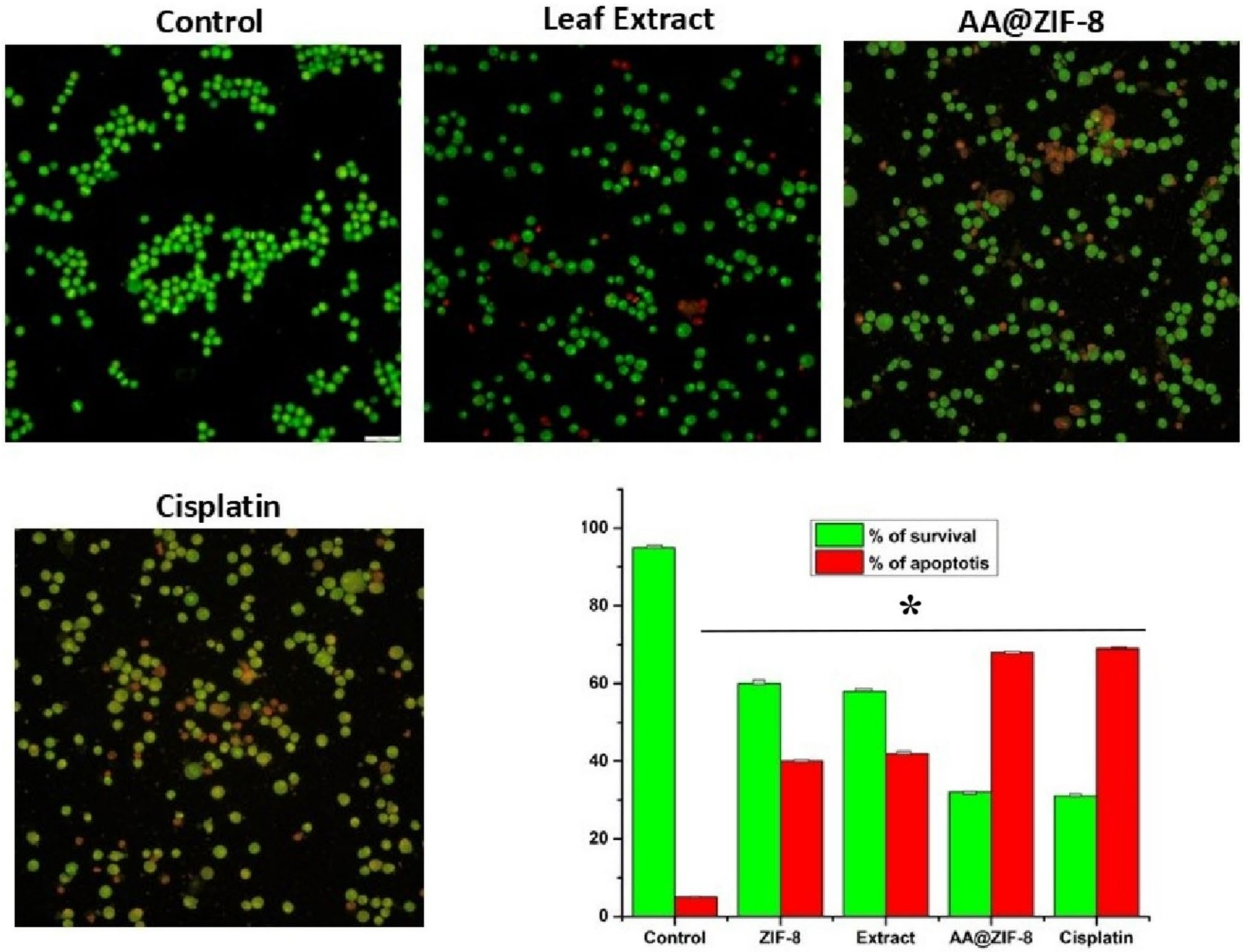

Fig. 5 Apoptotic effect of AA@ZIF-8 nanocomposite against A549 cells, bar graph represents the percentage of apoptotic and survival cells. $^*p < 0.05$ denotes the statistical significance between the control and treated groups

Results and discussion

Synthesis and characterization of theAA@ZIF-8 nanocomposite

The development of AA@ZIF-8 nanocomposite was confirmed by visually changing the color of the reaction mixture from white to pale green. In the one-pot synthesis technique, the methyl and hydroxyl groups in the bioactive molecule of *A. aspera* extract worked as a reducing agent, promoting the creation of ZIF-8. UV–VIS spectroscopy confirmed the formation of the AA@ZIF-8 nanocomposite (Fig. 1).

UV–vis spectroscopy analysis

A. aspera extract-loaded ZIF-8 nanoparticles have been successfully fabricated by a one-pot synthesis approach. Zinc ions and linker molecules are self-assembled by coordination bonds. The coordinated zinc ions and plant extract in AA@ZIF-8 nanocomposite were evaluated by UV –vis spectroscopy (Fig. 1A, B). There is no unidentified absorption peak in ZIF-8 nanoparticles. The bare ZIF-8 exhibits a peak at 228 nm. When plant extract was incorporated with AA@ZIF-8 nanocomposite exhibited alternative absorption peaks at 348 nm. It confirmed the intermolecular bonding between the plant extract and 2-methylimidazole organic linkers (Raju et al. 2023).

FTIR analysis

FTIR confirmed functional groups of ZIF-8, and AA@ZIF-8 nanocomposite as shown in Fig. 1 C-D. A broad absorption peak at 3321 cm^{-1} corresponds to a hydroxyl group (O–H) of water molecules. A sharp peak in AA@ZIF-8 nanocomposite at 3353 cm^{-1} is attributed to the

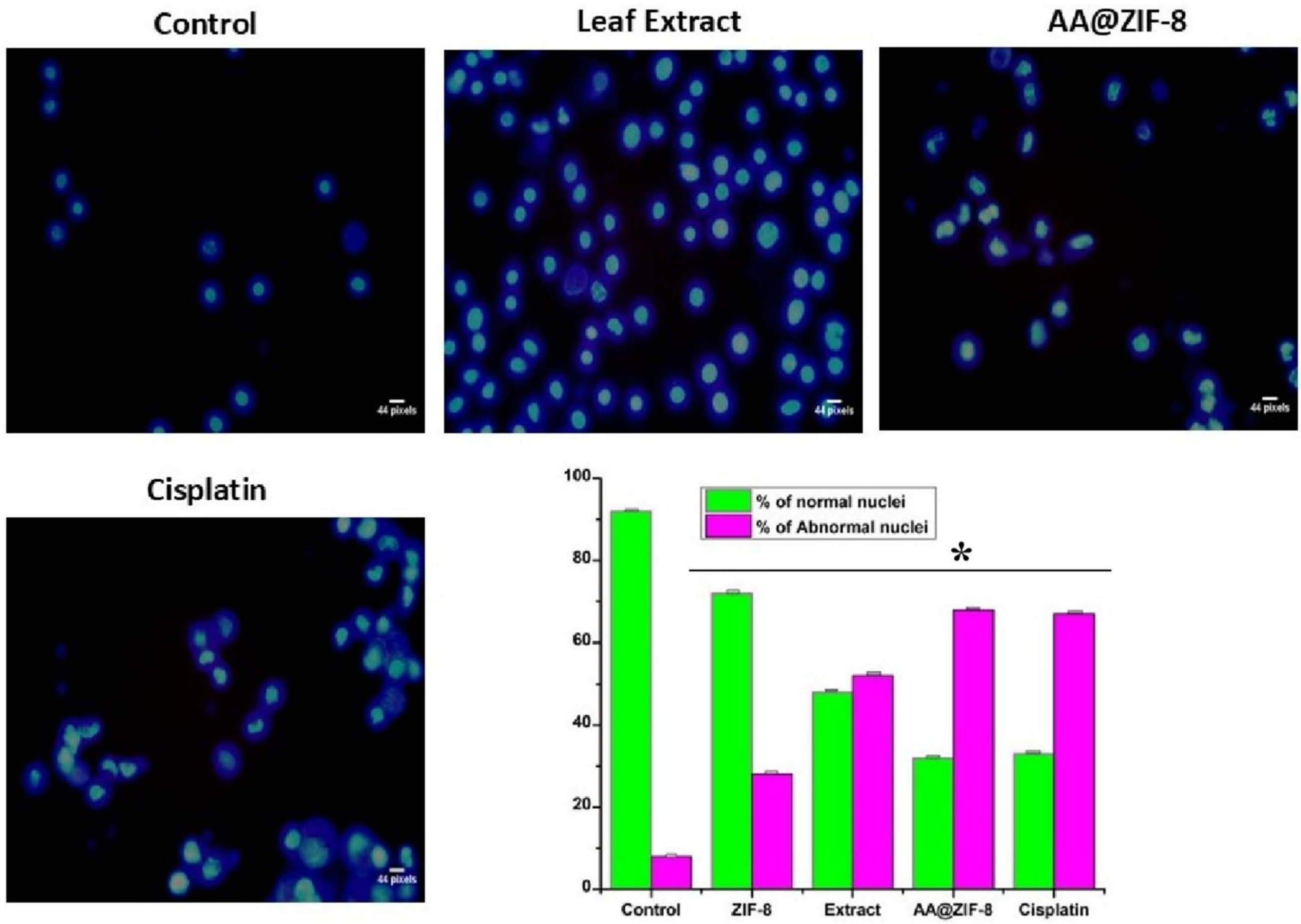

Fig. 6 ROS-mediated cell death of A549 cells after treatment with AA@ZIF-8 nanocomposite in DCFDA fluorescence assay. *$p < 0.05$ denotes the statistical significance between the control and treated groups

NH_2 amino groups. The number and positions of the IR spectrum in ZIF-8 and AA@ZIF-8 nanocomposite appeared to be very similar. The stretching vibration at 2931 cm^{-1} represents the presence of C-H and CH_3 groups (Prabhu et al. 2019). Aliphatic compounds (methyl C – H) in AA@ZIF-8 appear between 2950 and 2970 cm^{-1}. The peak at 1570 cm^{-1} contributes an aromatic stretching vibration of C–C – C in AA@ZIF-8 nanocomposite. The peak at 760 cm^{-1} confirms aromatic C – H stretching. In specific peak at around 420 cm^{-1} corresponded to Zn – N metal–organic linker 2-methyl imidazole in the AA@ZIF-8 nanocomposite (Raju et al. 2023).

XRD and DLS analysis

Powder XRD results revealed the high crystallinity of AA@ZIF-8 nanocomposite. The broadening of the AA@ZIF-8 peaks is because of the loading of extract within the ZIF-8 crystals (Fig. 2A). The diffraction peaks of AA@ZIF-8 and ZIF-8 matched with the bare ZIF-8 nanoparticles (JCPDS 00–062-1030) cubic crystalline structure. The diffraction spectrum of bare ZIF-8 and AA@ZIF-8 nanocomposite at $2\theta = 10.4°$, 12.62°, 16.35°, 18.03°, 19.42°, 24.5°, 25.4°, 29.66°, 31.45°, and 32.37°, which corresponds to the crystal planes of (011), (002), (112), (022), (013), (222), (114), (233), (134), and (044), respectively. In dynamic light scattering analysis, the average hydrodynamic size of AA@ZIF-8 nanocomposite indicates 100–120 nm. This size is highly suitable for cellular internalization and targeted drug delivery techniques (Fig. 2C). The size uniformity of AA@ZIF-8 nanocomposite in liquid medium indicates that there is no aggregation in liquid medium.

FE-SEM analysis

Surface nature of the synthesized AA@ZIF-8 nanocomposite was examined using the FE-SEM imaging method as shown in Fig. 3. The AA@ZIF-8 nanocomposite exhibited

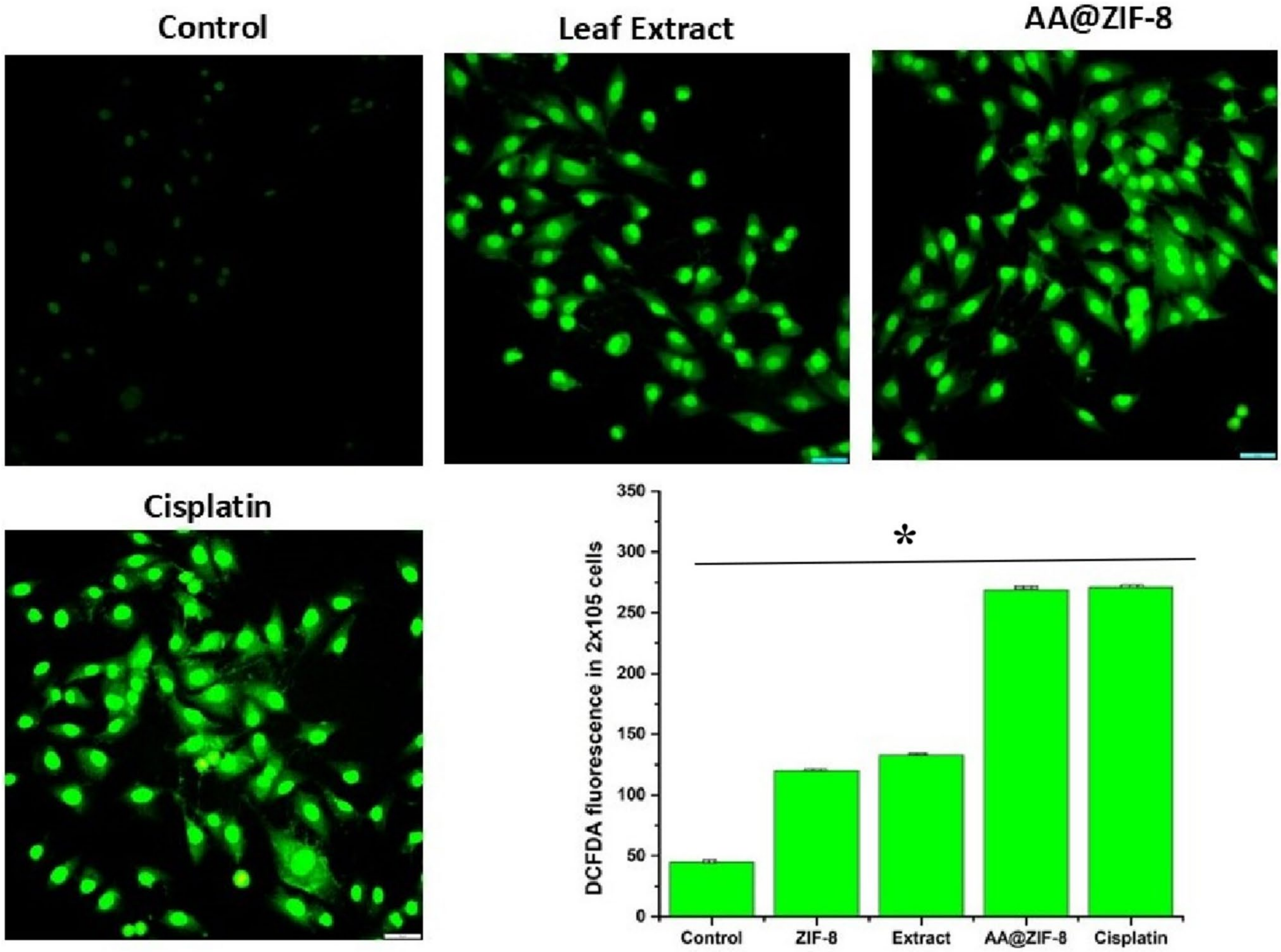

Fig. 7 Nuclear damage of A549 cells after treatment of AA@ZIF-8 nanocomposite on Hoechst 33,342 staining. Bar diagram illustrating the percentage of cells with nuclear damage. $*p < 0.05$ denotes the statistical significance between the control and treated groups

a crystalline size range of 120–240 nm. The *Achyranthes aspera* extract loaded AA@ZIF-8 showed a diameter range of 50–120 nm. The results of the SEM analysis confirmed that it is more suitable for biological applications (Rahimi et al. 2024). In SEM images of bare ZIF-8 nanoparticles show a uniform and soft surface morphology.

In vitro pH-responsive drug release mechanism of AA@ZIF-8 nanocomposite

The effect of *Achyranthes aspera* extract release in phosphate buffer saline, physiological, and acidic environments was investigated. The maximum extract loading (74%) is seen in the AA@ZIF-8 nanocomposite. In acidic conditions (pH ~ 5), the chitosan biopolymer surface coating significantly improves extract release in a stimulus-responsive manner, as shown in Fig. 2D. It is three times higher than the typical pH of 7.4 throughout the same period. Cleavage of metal–organic crosslinking results in the instantaneous release of extract at acidic pH = 5 and 6. When compared to acidic circumstances, the drug release percentage of AA@ZIF-8 nanocomposite at pH ~ 7.4 is extremely low. In physiological settings, the AA@ZIF-8 nanocomposite demonstrated a relatively modest release of around 19% throughout the 60 h. In this example, in acidic conditions, AA@ZIF-8 nanocomposite demonstrates 62% at pH-6, and 78% at pH-5 in 60 h. The results show that extract release decreases at pH = 7.4, whereas immediate release occurs at pH 5 and 6. It is an appropriate substance for controlling cancer cell proliferation that responds to stimuli.

Cytotoxicity

The unique biological characteristics of ZIF-8 nanoparticles have garnered increased attention in the biomedical sector. Nanocomposites are used as an effective material for agents in bioimaging, cancer treatment, and the prevention of microbial infections. Due to cytotoxicity and chemical

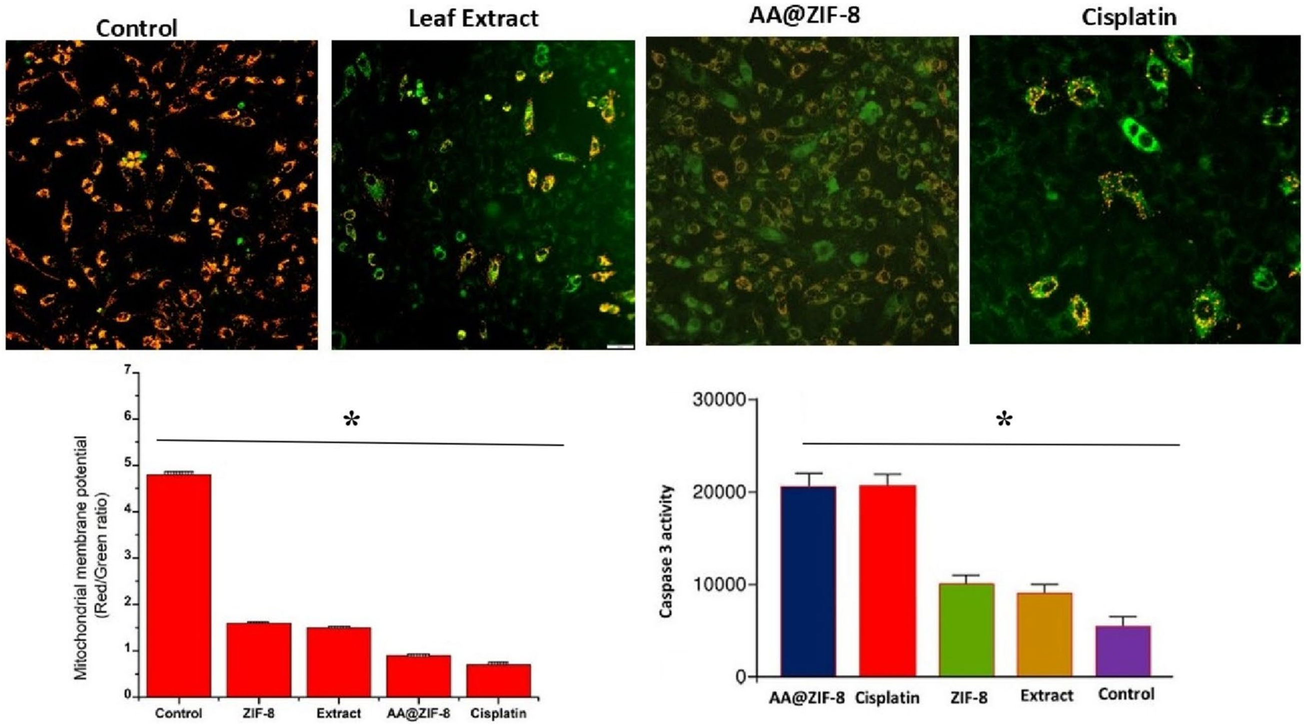

Fig. 8 Mitochondrial damage of A549 lung cancer cells after treatment with AA@ZIF-8 nanocomposite and Caspase 3 activity of AA@ZIF-8 nanocomposite. *$p < 0.05$ denotes the statistical significance between the control and treated groups

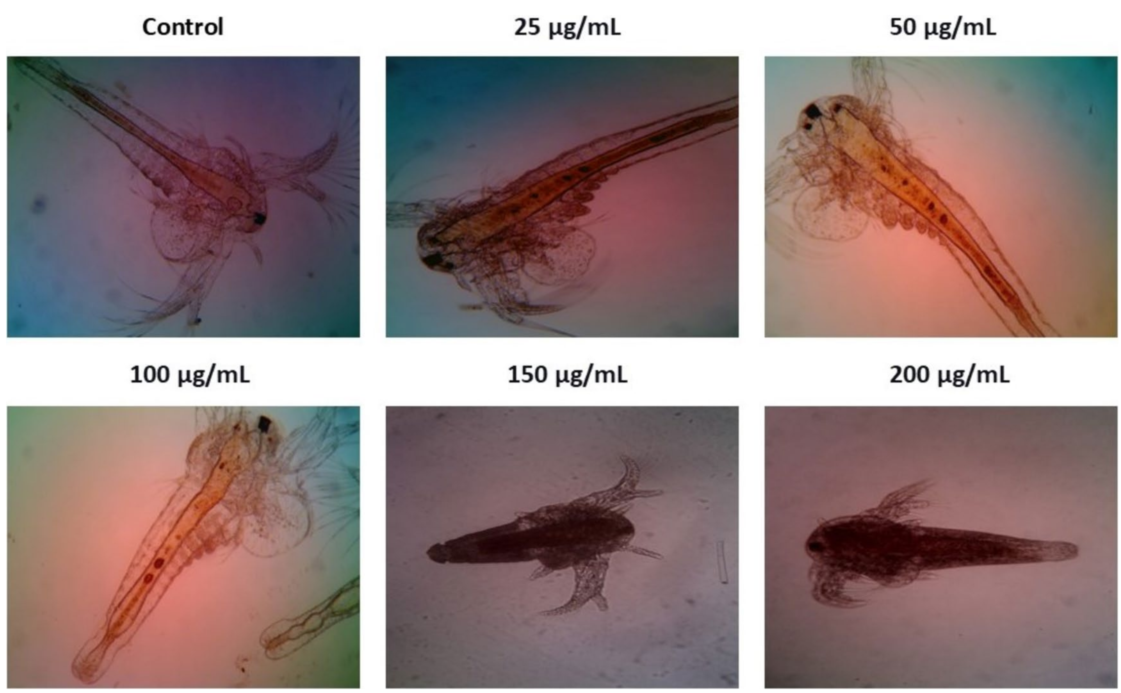

Fig. 9 In vivo toxic effect of AA@ZIF-8 against the *Artemia salina* model organism under an inverted phase contrast microscope at 10 × magnification

composition becomes more challenging for human usage. The accumulation of biomolecules in humans causes various health problems. Here, the cytotoxicity effect of AA@ZIF-8 nanocomposite against A549 cells was assessed by MTT assay. Cytotoxicity results suggested that it has excellent toxic effects against human lung cancer cells. IC_{50} values were observed to be 63.22 ± 0.03 µg/mL. The overall cytotoxicity studies revealed that AA@ZIF-8 nanocomposite is a suitable material for lung cancer treatment (Fig. 4). In this case, the IC_{50} concentration of *Achyranthes aspera* leaf extract against lung cancer cells in previous reports shows 79.78 ± 1.11 µg/mL (Omidiani et al. 2020). Our AA@ZIF-8 shows excellent cytotoxic effects at lower concentrations. Compared to existing reports, the zeolitic imidazole frameworks mediated AA@ZIF-8 nanocomposite is highly efficient against A549 lung cancer cells (Fig. 5).

ROS-mediated anticancer potential of AA@ZIF-8 nanocomposite

The AA@ZF-8 nanocomposite induces cell death via ROS generation in A549 cells after treatment. The DCFH/DA fluorescent intensity was raised after nanocomposite exposure, which indicates the overproduction of ROS in A549 cells. The quantitative results suggested that apoptosis depends upon the overproduction of ROS in the intracellular region. In the treated and control groups, there was no ROS production detectable at the initial time. After the AA@ZIF-8 nanocomposite exposure, a significant amount of ROS production appeared at 30 min (Fig. 6). The microscopic images of nanocomposite-treated groups showed a bright green fluorescence. The AA@ZIF-8 nanocomposite-mediated ROS production has a close relationship with mitochondrial dysfunction and cell death mechanisms. The caspase-3 activation was evaluated in A549 cells treated with the AA@ZIF-8 nanocomposite (Fig. 8). Colorimetric assay of caspase-3/9 activity showing that the caspase activity in cells treated with nanocomposite significantly increases in comparison to the untreated groups. The total percentage of caspase-3 activity is twofold higher than alone leaf extract alone and ZIF-8 nanoparticles. The results indicated that AA@ZIF-8 nanocomposite promoted the caspase-3/9 activation in A549 cells. It is a key factor in the apoptotic mechanism of lung cancer cells.

Assessment of the apoptotic effect of AA@ZIF-8 against A549 lung cancer cells

The apoptosis mechanism of AA@ZIF-8 nanocomposite post exposed cell was observed by AO/Br dual staining. Treatment of AA@ZIF-8 nanocomposite leads the structural changes in A549 cells. The control cells do not show any nuclear deformities. The changes in nuclear and cell wall region were found in A549 cells after the nanocomposite treatment. Apoptotic percentage was quantified by an AO/EB staining method (Fig. 5). In the AO/Br staining normal cells are emitting bright green fluorescence and can be distinguished from apoptosis. After the AA@ZIF-8 nanocomposite treatment, a notable level of death cell was detected in nanocomposite exposed groups. The previous research confirmed that, extract of *Achyranthes aspera* induces apoptosis in cancer cells by triggering the mitochondrial pathway, activating caspase-3, and potentially inhibiting the PKCα signaling pathway. This AO/EB dual staining results suggested that A549 cell death after AA@ZIF-8 nanocomposite treatment occurred through the apoptotic process.

Assessment of nuclear damage of AA@ZIF-8 against A549 cells

The nuclear changes of A549 cells after treatment with AA@ZIF-8 were examined by the Hoechst 33,342 staining assay. The percentage of normal and abnormal cell nuclei of the AA@ZIF-8 treated group showed 70.35 ± 0.13% of abnormal nuclei. In this case, control cells exhibited only 4.1 ± 0.15% abnormal cell nuclei. Fluorescence microscopic image of control cells showed a maximum amount of normal cell nuclei. In this case, AA@ZIF-8 nanocomposite-treated cells showed bright blue fluorescence intensity with cellular modifications such as nuclear irregularity, fragmented chromatins, and shrinkages (Fig. 7). The cellular changes suggested the fact that apoptosis led to cell death. The death of A549 cells is reflected by the anticancer efficacy of AA@ZIF-8 nanocomposite.

JC-1 staining

In JC-1 staining after AA@ZIF-8 nanocomposite exposure, A549 cells showed excellent mitochondrial damage-mediated cell death. The percentage of apoptotic cells was multiplied based on the time duration and the dose of nanocomposite. The mitochondrial damage mechanism is a leading mode of action for apoptosis. The AA@ZIF-8 nanocomposite induced the caspase-3 pathway, leading to cell death in A549 cells. The decrease in mitochondrial membrane integrity prompted the apoptosis process (Fig. 8). More than 50% of treated cells showed mitochondrial membrane damage.

Biocompatibility

To ensure biocompatibility, nanomaterials for biomedical purposes must undergo safety assessments before usage

(Tiwari et al. 2017). Biocompatibility of the AA@ZIF-8 nanocomposite was investigated against the brine shrimp *Artemia salina* animal model. The brine shrimp acute toxicity test is a simple, inexpensive, and reliable method for animal toxicological studies (Prabhu et al. 2019). The LC_{50} value of the AA@ZIF-8 nanocomposite was quantified to be 160.13 ± 2.82 µg/mL. The morphological alteration of treated *Artemia salina* nauplii was examined under a phase contrast microscope (Fig. 9). The images of treated nauplii showed no structural changes and growth reduction up to the concentration of 150 µg/mL. The size reduction and minimum number of dead nauplii have appeared at the highest concentration of 175 µg/mL. The results indicate that AA@ZIF-8 nanocomposite showed low cytotoxic effects until 175 µg/mL concentration. No other swimming behavior changes were observed until 48 h. The synthesized AA@ZIF-8 nanocomposite is biosafe at lower concentrations and mildly toxic at higher doses. In contrast, the AA@ZIF-8 nanocomposite caused no discernible disruption in non-cancer cells or membranes (Tong et al. 2013; Rajabi et al. 2015). This toxicity study confirmed that AA@ZIF-8 nanocomposite is biocompatible for anticancer applications.

Conclusion

In this study, we fabricated a bioactive *Achyranthes aspera* leaf extract loaded with ZIF-8 nanocomposite for improved wound healing and anticancer applications. This nanocomposite has the potential to release phytocompounds and zinc ions in the targeted cell sites due to the intracellular caspase-3 and apoptotic mechanism. The results of biocompatibility studies suggested, there were no other toxic effects observed against the *Artemia salina* animal model. The anticancer activity results suggested that AA@ZIF-8 leads to nuclear dysfunction and ROS-mediated cytotoxicity against A549 cells. The bioactive potential of AA@ZIF-8 nanocomposite causes excellent cytotoxic efficacy in lower concentrations. The overall experimental findings strongly indicated that the AA@ZIF-8 nanocomposite is a potential biomaterial for anticancer applications. Nanocomposite might be a promising, low-cost, and biosafe material for cancer therapies, thereby lowering treatment costs.

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Data availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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